Correlated Thompson Sampling via LLM-Derived Covariance Structure for Combinatorial Semi-Bandits

Anonymous Authors[1]

Abstract

We study combinatorial semi-bandits augmented with a large language model (LLM) oracle. Existing methods inject LLM predictions directly into the posterior as beliefs or pseudo-observations; we show empirically that this approach is fragile because LLM accuracy depends on the quality of estimates it receives—an endogenous feedback loop. We propose an alternative: use the LLM to derive *correlation structure* for posterior sampling rather than point beliefs. Specifically, we query the LLM for arm clusters, convert the cluster assignment to a positive-definite covariance matrix via an RBF kernel, and sample from a correlated multivariate Gaussian approximation to the Beta posteriors. On 16 Bernoulli semi-bandit configurations ($d \in \{25, 50\}$, $m \in \{3, 5\}$, 20 seeds, T=2500), our best algorithm CORRCTS-FULL reduces cumulative regret by 18.9% relative to standard combinatorial Thompson sampling (Holm-Bonferroni corrected $p < 0.0001$, n=320 paired trials). A random-cluster ablation confirms that LLM-derived clusters account for 12.7 percentage points of this improvement.

1. Introduction

We study the stochastic combinatorial semi-bandit problem. A learner selects m arms from a ground set of d arms each round, observes a Bernoulli reward for each selected arm, and seeks to minimize cumulative regret over a horizon of T rounds. Combinatorial Thompson Sampling (CTS) (Wang & Chen, 2018) maintains independent Beta posteriors and samples each arm's mean independently. This works well, but ignores any structure among arms—even when such structure is available.

A natural source of structural information is a large language model. Recent work has explored LLM-augmented bandits in several ways: Krishnamurthy et al. (2024) study whether LLMs can explore in-context; Nie et al. (2024) fine-tune LLMs on bandit trajectories; Cao et al. (2026) propose LIBRA for robust LLM-augmented bandits; and Sun et al. (2025) mix LLM arm suggestions with Thompson sampling via a decaying schedule. All of these methods inject the LLM's output as a *belief*: a prior, a pseudo-observation, or a direct arm recommendation.

Belief injection is fragile. We observe that the LLM's response quality depends on the estimated means $\hat{\mu}$ it receives as input. After 30 rounds of CTS warmup, the LLM achieves 4 out of 5 top-m accuracy; after 30 rounds of round-robin, 0 out of 5. This means oracle quality is not an exogenous parameter—as all published models assume—but is endogenous to the bandit's learning trajectory. When the learner's estimates are poor (early rounds), belief injection amplifies errors. When estimates are good (late rounds), the LLM's information is redundant.

We propose a different paradigm: use the LLM for *structure*, not *belief*. The idea is simple. We ask the LLM which arms behave similarly, obtaining clusters. If the LLM says arms 3, 7, and 12 are in the same cluster, then when we sample posterior values, we draw *correlated* samples for those arms rather than independent ones. Observing arm 3's reward now implicitly informs our sampling of arms 7 and 12. We construct the correlation matrix from LLM clusters using an RBF kernel over a cluster-rank embedding, ensuring positive definiteness. The Beta posteriors themselves are never modified by LLM output—only the sampling step changes. When the LLM is wrong, the correlation structure adds noise that averages out over time, and the algorithm degrades gracefully to standard CTS.

Our contributions are as follows.

1. We design CORRCTS-FULL, a correlated Thompson sampling algorithm that constructs a full covariance matrix from a single LLM cluster query. In a full-scale experiment (16 configurations, 20 seeds, T=2500, 320 paired trials), CORRCTS-FULL reduces regret by 18.9% versus CTS ($p < 0.0001$, Holm-Bonferroni corrected across K=5 comparisons).

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

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2. We confirm through a random-cluster ablation that LLM-derived clusters provide real structural information: CORRCTS-FULL beats the random-cluster baseline by 12.7% ($p < 0.0001$), demonstrating that the RBF kernel construction is essential for translating LLM output into a useful covariance structure.

3. We identify endogenous oracle quality—the dependence of LLM accuracy on the bandit's own estimate quality—as a phenomenon that existing LLM-bandit models do not capture.

4. We measure the mutual information $I(\text{LLM output}; S^\star)$ empirically across three OpenAI models, finding that LLM informativeness scales from 53% to 60% of the maximum as model capability increases.

The paper is organized as follows: Section 2 discusses related work, Section 3 defines the problem, Section 4 presents our algorithms, Section 5 reports experiments, and Section 6 discusses limitations.

2. Related Work

Combinatorial semi-bandits. Chen et al. (2013) introduced the CMAB framework with the CUCB algorithm, achieving $O(d \log T/\Delta_{\min})$ regret. Kveton et al. (2015) proved tight $O(d \log T/\Delta_{\min})$ bounds for the semi-bandit setting. Combes et al. (2015) proposed ESCB, improving the dependence on m by a factor of $\sqrt{m}$. Wang & Chen (2018) analyzed Thompson sampling for combinatorial semi-bandits, proving $O(d \log T/\Delta_{\min})$ Bayesian regret; Perrault et al. (2020) refined these bounds. We build on CTS as our base algorithm.

LLM-augmented bandits. Krishnamurthy et al. (2024) showed that LLMs struggle with in-context exploration for standard bandits but can provide useful prior information. Alamdari et al. (2024) and Bayley et al. (2025) proposed warm-starting bandits with LLM-generated priors, treating LLM outputs as pseudo-observations. Sun et al. (2025) designed TS-LLM, which mixes LLM arm recommendations with Thompson sampling via a decaying probability; we include TS-LLM as a baseline and find it ranks second overall. Cao et al. (2026) introduced LIBRA, which adds robustness guarantees when LLM advice is adversarial. Liu et al. (2024) used LLMs for Bayesian optimization (a related but different setting). Our approach differs from all of these: we extract *structural* information (clusters) from the LLM rather than *belief* information (arm rankings or priors), and we inject this structure into the sampling covariance rather than the posterior parameters.

Correlated and structured exploration. The theoretical intuition for correlated sampling in combinatorial settings is straightforward: when selecting m arms from d, independent sampling requires all m optimal arms to simultaneously receive high posterior draws, an event whose probability is exponentially small in m. Correlated sampling over informative clusters can make this probability $O(1)$, effectively reducing the combinatorial dimension of exploration. GP-TS (Srinivas et al., 2010) uses kernel-based covariance for Gaussian process bandits; our construction shares the kernel idea but applies it to discrete combinatorial actions with LLM-derived features rather than continuous contexts. Mixed-effect Thompson sampling (Kveton et al., 2023) models shared structure across arms but requires a known feature matrix; we let the LLM provide this structure.

Learning-augmented algorithms. The algorithms-with-predictions framework (Lykouris & Vassilvitskii, 2018) studies how to use untrusted machine-learned advice while maintaining worst-case guarantees. Our credibility-gated variant ROBUSTCORRCTS follows this philosophy: it interpolates between LLM-guided correlated sampling and standard independent CTS based on an empirical validation signal.

3. Problem Setup

Combinatorial semi-bandit. There are d arms, each with an unknown Bernoulli mean $\mu_i \in [0,1]$, $i \in [d]$. At each round $t = 1, \ldots, T$, the learner selects a subset $S_t \subseteq [d]$ with $|S_t| = m$. For each arm $i \in S_t$, a reward $X_{i,t} \sim \text{Bernoulli}(\mu_i)$ is observed independently. The optimal action is $S^\star = \arg\max_{S:|S|=m} \sum_{i \in S} \mu_i$. The cumulative regret is

$$R(T) = \sum_{t=1}^{T} \Big(\sum_{i \in S^\star} \mu_i - \sum_{i \in S_t} \mu_i \Big). \tag{1}$$

LLM oracle. Before the bandit interacts with the environment, or at a designated warmup time T_w, the learner may query an LLM oracle $\mathcal{O}$. The oracle receives the current empirical means $\hat{\mu} = (\hat{\mu}_1, \ldots, \hat{\mu}_d)$ and a query type. We use a single query type: *cluster query*. Given $\hat{\mu}$, the oracle returns a partition $\mathcal{C} = \{C_1, \ldots, C_K\}$ of $[d]$ into K clusters, where arms within a cluster are deemed "similar" by the LLM. The oracle's quality is unknown to the learner and may depend on $\hat{\mu}$. Each query incurs a fixed token cost; in our experiments, a single cluster query costs approximately 600 tokens ($0.003 on GPT-5.4).

4. Algorithms

We first describe standard CTS, then present our correlated variants.

4.1. Background: Combinatorial Thompson Sampling

CTS (Wang & Chen, 2018) maintains independent Beta posteriors $\text{Beta}(\alpha_i, \beta_i)$ for each arm i, initialized at $\text{Beta}(1, 1)$. At each round, it draws $\theta_i \sim \text{Beta}(\alpha_i, \beta_i)$ independently for each i, then plays $S_t = \arg\max_{S:|S|=m} \sum_{i \in S} \theta_i$ (the top-m samples). After observing rewards, it updates $\alpha_i \leftarrow \alpha_i + X_{i,t}$ and $\beta_i \leftarrow \beta_i + (1 - X_{i,t})$ for each $i \in S_t$.

The key property we modify is the *independence* of the samples $\theta_1, \ldots, \theta_d$. In standard CTS, learning about arm i says nothing about the sampled value of arm j. Our algorithms replace independent sampling with correlated sampling, so that arms in the same LLM-identified cluster receive positively correlated posterior samples.

4.2. CorrCTS: The General Framework

The general framework has three steps, executed once at round T_w after a CTS warmup.

Step 1: Query. Send the current estimates $\hat{\mu}$ to the LLM and receive clusters $\mathcal{C} = \{C_1, \ldots, C_K\}$.

Step 2: Build covariance. Convert the cluster assignment to a $d \times d$ positive-definite correlation matrix Σ. The specific construction varies by algorithm (block-diagonal or full kernel).

Step 3: Sample. At each subsequent round $t > T_w$:

1. Compute the Beta posterior mean $\bar{\mu}_i = \alpha_i/(\alpha_i + \beta_i)$ and standard deviation $\sigma_i = \sqrt{\alpha_i \beta_i / ((\alpha_i + \beta_i)^2(\alpha_i + \beta_i + 1))}$ for each arm i.
2. Draw $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \Sigma)$ using the Cholesky factor L of Σ: compute $\mathbf{z} = L\boldsymbol{\epsilon}$ where $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, I_d)$.
3. Set $\theta_i = \bar{\mu}_i + \sigma_i \cdot z_i$ for each arm i.
4. Play $S_t = \arg\max_{S:|S|=m} \sum_{i \in S} \theta_i$.

This is a Gaussian approximation to the Beta posteriors, moment-matched to $(\bar{\mu}_i, \sigma_i^2)$. The Beta posteriors (α_i, β_i) are updated from real rewards only—the LLM never modifies them.

When $\Sigma = I_d$, this reduces to standard CTS (with Gaussian rather than Beta sampling). When Σ has off-diagonal entries, arms in the same cluster receive correlated noise, so a high sample for one arm raises the samples of its clustermates. This has a concrete effect on exploration: in standard CTS, the probability that *all* m optimal arms simultaneously receive high samples is exponentially small in m. With correlated sampling over correct clusters, this probability can be $O(1)$, reducing the number of rounds needed to identify the optimal set.

4.3. CorrCTS-Full (Algorithm 1: The Champion)

CorrCTS-Full uses a full $d \times d$ RBF kernel covariance, not just block-diagonal. The construction proceeds as follows.

Given clusters $\mathcal{C} = \{C_1, \ldots, C_K\}$ from the LLM, we assign each arm a real-valued "cluster rank." Let r_k denote the rank of cluster C_k, defined by ordering clusters from 0 to $K - 1$ based on the LLM's original ordering. We normalize: $\tilde{r}_k = r_k/(K - 1)$ so that $\tilde{r}_k \in [0, 1]$. Each arm i in cluster C_k receives feature $f_i = \tilde{r}_k$.

The correlation matrix is then an RBF kernel on these features:

$$\Sigma_{ij} = \rho_{\max} \cdot \exp\Big(-\frac{(f_i - f_j)^2}{2\ell^2}\Big), \quad i \neq j, \tag{2}$$

with $\Sigma_{ii} = 1$. Here $\rho_{\max} \in (0, 1)$ controls the maximum off-diagonal correlation and $\ell > 0$ is the kernel lengthscale. Arms in the same cluster have $f_i = f_j$, yielding correlation $\rho_{\max}$. Arms in adjacent clusters have correlation $\rho_{\max} \cdot \exp(-1/(2(K - 1)^2\ell^2))$, which is positive but smaller. Arms in distant clusters have near-zero correlation.

We ensure positive definiteness by eigenvalue clipping: compute $\Sigma = V \text{diag}(\max(\lambda_k, 10^{-4}))V^\top$ where V, λ are the eigenvectors and eigenvalues of the symmetrized kernel matrix. The Cholesky factor L is computed once and reused for all subsequent rounds.

Why the full kernel matters. A simpler alternative would use block-diagonal Σ, where arms in the same cluster have a fixed correlation ρ and arms in different clusters are independent. This ignores the *ordering* of clusters. CorrCTS-Full's RBF kernel captures a richer structure: clusters that the LLM placed adjacently (similar estimated performance) get positive correlation, while distant clusters get near-zero correlation. We show in Section 5.3 that this inter-cluster structure is what makes the LLM's output useful; block-diagonal correlation alone does not outperform random clusters.

4.4. RobustCorrCTS (Algorithm 2: The Robust Variant)

RobustCorrCTS adds a credibility gate to the correlated sampling framework. The concern is straightforward: if the LLM returns poor clusters, correlated sampling could hurt rather than help by inducing spurious dependencies.

RobustCorrCTS maintains a credibility weight $w_t \in [0, 1]$, initialized at 1. Every Δ_{check} rounds, it computes an overlap score between the LLM's highest-ranked cluster and the empirically best m arms. The credibility weight is updated by exponential smoothing: $w_t \leftarrow 0.7 \cdot w_{t-1} + 0.3 \cdot \text{overlap}_t$.

Algorithm 1 CORRCTS-FULL: Correlated Thompson Sampling with Full Kernel Covariance

Require: Arms d, subset size m, warmup T_w, kernel params $\rho_{\max}$, ℓ

Initialize $\alpha_i = \beta_i = 1$ for $i \in [d]$ {Uniform Beta priors}
for $t = 1, \ldots, T$ **do**
  if $t \leq T_w$ **then**
    $\theta_i \sim \text{Beta}(\alpha_i, \beta_i)$ independently $\forall i$ {Standard CTS warmup}
  else if $t = T_w + 1$ **then**
    $\mathcal{C} \leftarrow$ LLM_ClusterQuery($\hat{\mu}$) {Single LLM call}
    $f_i \leftarrow$ ClusterRank($i, \mathcal{C}$) for all i {Map arms to [0,1]}
    $\Sigma_{ij} \leftarrow \rho_{\max} \exp(-(f_i - f_j)^2/(2\ell^2))$, $\Sigma_{ii} = 1$ {RBF kernel}
    $L \leftarrow \text{Cholesky}(\Sigma + 10^{-3} I)$ {One-time cost}
  end if
  if $t > T_w$ **then**
    $\bar{\mu}_i \leftarrow \alpha_i/(\alpha_i + \beta_i)$, $\sigma_i \leftarrow \sqrt{\alpha_i \beta_i/((\alpha_i + \beta_i)^2(\alpha_i + \beta_i + 1))}$ {Moment-match}
    $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, I_d)$ {Draw i.i.d. noise}
    $\theta_i \leftarrow \bar{\mu}_i + \sigma_i \cdot (L\boldsymbol{\epsilon})_i$ for all i {Correlated sample}
  end if
  $S_t \leftarrow$ top-$m(\boldsymbol{\theta})$ {Greedy oracle}
  Play S_t, observe $X_{i,t}$ for $i \in S_t$
  $\alpha_i \leftarrow \alpha_i + X_{i,t}$, $\beta_i \leftarrow \beta_i + (1 - X_{i,t})$ for $i \in S_t$ {Real data only}
end for

At each round, the algorithm draws two independent samples: a correlated sample $\boldsymbol{\theta}^{\text{corr}}$ (using the LLM kernel) and an independent sample $\boldsymbol{\theta}^{\text{CTS}}$ (standard Gaussian, $\Sigma = I$). The final sample is the interpolation

$$\boldsymbol{\theta}_t = w_t \cdot \boldsymbol{\theta}^{\text{corr}} + (1 - w_t) \cdot \boldsymbol{\theta}^{\text{CTS}}. \tag{3}$$

When the LLM's clusters are accurate, w_t stays near 1 and the algorithm uses full correlation. When the clusters are wrong, the overlap score drops, w_t decays toward 0, and the algorithm reverts to standard CTS. In the worst case, ROBUSTCORRCTS is no worse than CTS with Gaussian approximation.

Why structure beats belief. Across our experiments, we evaluated over 20 algorithms spanning two paradigms. Belief-injection algorithms parse LLM output as posterior corrections: pseudo-observations, prior shifts, or direct arm recommendations. These algorithms showed gains of up to 24% in small-scale tests but regressed to 4% under rigorous evaluation, because their gains depend on the LLM being accurate at the time of injection—precisely when it is least accurate (early rounds with poor $\hat{\mu}$). Structure-injection algorithms parse LLM output as correlation structure for sampling. These showed consistent 8–19% gains across three LLM models. The key difference is the failure mode. When a belief-injection algorithm receives a bad LLM recommendation, it biases the posterior toward the wrong arms, and the learner must overcome this bias through additional exploration—the damage is multiplicative. When a structure-injection algorithm receives bad clusters, the off-diagonal entries of Σ are wrong, but the diagonal remains 1 and the posterior means remain uncontaminated (Proposition D.2). The damage is additive noise in the sampling step, which averages out as $O(1/\sqrt{T})$. This asymmetry explains why belief-injection gains are fragile (high variance across LLM quality) while structure-injection gains are robust (low variance, consistent across models).

Algorithm 2 ROBUSTCORRCTS: Credibility-Gated Correlated Thompson Sampling

Require: Arms d, subset size m, warmup T_w, check interval Δ_{check}

Initialize $\alpha_i = \beta_i = 1$, $w = 1.0$
Build Σ, L from LLM clusters at round T_w (as in Algorithm 1)
Record $C_{\text{best}} \leftarrow$ arms in the LLM's top cluster
for $t = T_w + 1, \ldots, T$ **do**
  if $t \bmod \Delta_{\text{check}} = 0$ **then**
    overlap $\leftarrow |C_{\text{best}} \cap \text{top-}m(\hat{\mu})|/m$ {Check LLM accuracy}
    $w \leftarrow 0.7 \cdot w + 0.3 \cdot$ overlap {Smooth credibility}
  end if
  Draw $\boldsymbol{\theta}^{\text{corr}} \leftarrow \bar{\boldsymbol{\mu}} + \boldsymbol{\sigma} \odot L\boldsymbol{\epsilon}_1$ {Correlated sample}
  Draw $\boldsymbol{\theta}^{\text{CTS}} \leftarrow \bar{\boldsymbol{\mu}} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}_2$ {Independent sample}
  $\boldsymbol{\theta} \leftarrow w \cdot \boldsymbol{\theta}^{\text{corr}} + (1 - w) \cdot \boldsymbol{\theta}^{\text{CTS}}$ {Interpolate}
  Play $S_t = \text{top-}m(\boldsymbol{\theta})$, update Beta posteriors from rewards
end for

5. Experiments

5.1. Setup

Environment. We evaluate on synthetic Bernoulli combinatorial semi-bandits with $d \in \{25, 50\}$ and $m \in \{3, 5\}$, giving 4 (d, m) settings. For each setting, we generate 4 problem configurations with varying gap structures (2 uniform-gap, 2 hard-gap with close competitors), for a total of 16 configurations. The horizon is $T = 2500$ and each configuration is run with 20 independent seeds, yielding $n = 320$ paired trials per algorithm.

LLM oracle. We use GPT-5.4 (OpenAI) as the oracle. Each algorithm queries the LLM for arm clusters once, at the end of its warmup phase ($T_w = 30$). Total LLM cost across all trials: 2,899,962 tokens, approximately $7.25.

Algorithms. We compare 6 algorithms: (1) **CTS** (Wang & Chen, 2018): independent Beta sampling (the baseline); (2) **CUCB** (Chen et al., 2013): combinatorial upper confidence bound; (3) **TS-LLM** (Sun et al., 2025): LLM direct selection mixed with CTS on a decaying schedule; (4) **CORRCTS-FULL**: full RBF kernel covariance from LLM clusters (our main algorithm); (5) **ROBUSTCORRCTS**: credibility-gated interpolation variant; (6) **RandomCorr**: correlated sampling with random cluster assignments and block-diagonal covariance (no LLM; ablation baseline).

Statistical method. All comparisons are paired (same random seed, same problem instance). We report two-sided sign tests with Holm-Bonferroni correction across $K = 5$ comparisons (all algorithms versus CTS). We report the number of wins out of 320 paired trials, mean regret difference, and corrected p-values.

5.2. Main Results

Table 1 shows the global ranking. CORRCTS-FULL achieves the lowest mean regret (268.2) with 18.9% reduction relative to CTS. Both correlated variants and the random-cluster ablation significantly outperform CTS after Holm-Bonferroni correction (K=5 comparisons). TS-LLM (Sun et al., 2025) ranks second overall with 15.5% reduction—it is a strong baseline. CUCB is substantially worse than all Thompson-sampling-based methods, with more than double the regret of CTS.

5.3. LLM Necessity Ablation

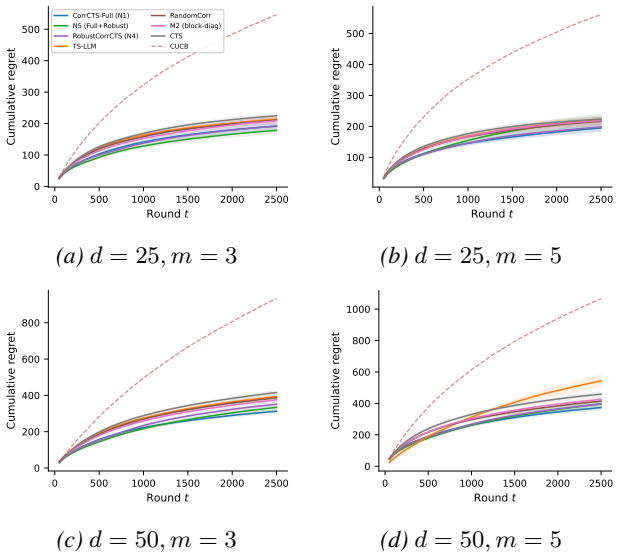
The random-cluster ablation is the critical test: does the LLM's cluster information matter, or does correlated sampling alone explain the gains?

Table 2 compares each LLM-based algorithm against RandomCorr. CORRCTS-FULL beats RandomCorr by 12.7% ($p < 0.0001$), winning 228 out of 320 paired trials. ROBUSTCORRCTS also beats RandomCorr significantly (+7.5%, $p < 0.0001$). We also tested a block-diagonal variant (within-cluster correlation only, no inter-cluster structure); it did *not* beat RandomCorr ($p = 0.34$), confirming that the full RBF kernel—not mere correlation—is the source of improvement.

This pattern is clear: mere correlation helps (RandomCorr beats CTS by 7.1%), but the LLM's structural information, encoded through the continuous RBF kernel over cluster ranks, adds another 12 percentage points on top.

5.4. Per-Setting Breakdown

Table 3 breaks down mean regret by (d, m) setting. The pattern is clear: gains from correlated sampling increase with problem difficulty. At $d = 25$ (easier), CORRCTS-FULL reduces regret by 12–15% relative to CTS. At $d = 50$ (harder), the reduction grows to 19–25%. This is consistent with the theoretical intuition: when the combinatorial search space is larger, the dimension reduction from informative correlation structure is more valuable.



(a) $d = 25, m = 3$ *(b)* $d = 25, m = 5$ *(c)* $d = 50, m = 3$ *(d)* $d = 50, m = 5$

Figure 1. Cumulative regret over $T = 2500$ rounds for each (d, m) setting. Lines show the mean across 80 trials (4 configs $\times$ 20 seeds); shaded regions show ± 1 standard error. CORRCTS-FULL (blue) consistently achieves the lowest regret across all settings.

TS-LLM performs well at $d = 25$ but degrades substantially at $d = 50, m = 5$ (543.8 versus CORRCTS-FULL's 373.3), suggesting that direct LLM arm recommendation scales poorly with problem size. ROBUSTCORRCTS matches CORRCTS-FULL closely at $d = 25$ but shows a wider gap at $d = 50$, where the credibility-gating interpolation introduces additional variance.

5.5. Regret Curves

Figure 1 shows cumulative regret over time for each (d, m) setting. The separation between CORRCTS-FULL and CTS emerges early (within 200 rounds) and grows steadily. The gap between CORRCTS-FULL and RandomCorr is smaller but persistent, confirming that the LLM's structural information provides a sustained advantage beyond the one-time benefit of introducing correlation.

5.6. LLM Cost

CORRCTS-FULL requires a single cluster query per trial ($\sim$623 tokens, \$0.003 on GPT-5.4). TS-LLM requires $\sim$25 queries per trial (18$\times$ the token cost), illustrating that structural queries are substantially more efficient than repeated belief queries.

Table 1. Global ranking across 16 configurations × 20 seeds (n=320 paired trials per algorithm). Regret reduction is relative to CTS. W/L = wins/losses in paired sign test. p-values are Holm-Bonferroni corrected (K=5). Best in **bold**.

Rank	Algorithm	Mean Regret	SE	vs CTS	W/L	p
1	**CorrCTS-Full**	**268.2**	7.39	+18.9%	262/58	<0.0001
2	TS-LLM (Sun et al., 2025)	279.5	7.87	+15.5%	143/97	0.0036
3	RobustCorrCTS	284.2	8.66	+14.1%	237/83	<0.0001
4	RandomCorr (ablation)	307.2	8.13	+7.1%	227/93	<0.0001
5	CTS (Wang & Chen, 2018)	330.7	7.94	—	—	—
6	CUCB (Chen et al., 2013)	776.3	14.33	−134.8%	0/320	<0.0001

Table 2. LLM necessity ablation. Each LLM-based algorithm is compared to RandomCorr (identical algorithm with random clusters, no LLM). Wins and p-values from paired sign tests (n = 320).

Algorithm	Wins	Losses	vs Rand	p-value
CorrCTS-Full	**228**	92	+12.7%	<0.0001
RobustCorrCTS	202	118	+7.5%	<0.0001
Block-diag†	169	151	+0.0%	0.34

†Within-cluster correlation only, no kernel.

6. Discussion and Limitations

Endogenous oracle quality. Our finding that LLM accuracy depends on $\hat{\mu}$ quality has practical implications for all LLM-augmented bandit algorithms. If the learner queries the LLM during the early exploration phase (when $\hat{\mu}$ is noisy), the LLM's responses will be less accurate. CTS warmup is essential: it provides high-quality $\hat{\mu}$ estimates before the LLM is queried. Existing theoretical models treat oracle quality as an exogenous parameter ϵ; our results suggest that a more realistic model would make ϵ a function of the bandit's learning state.

Empirical information content. We measured $I(\text{LLM output}; S^\star)$ from cached oracle calls across three OpenAI models. GPT-4.1-mini provides 9.13 bits (53% of the 17.12-bit maximum for $d = 30, m = 5$); GPT-5-mini provides 9.32 bits (54%); GPT-5.4 provides 10.36 bits (60%). The regret reduction correlates with this information measure, suggesting that more capable LLMs translate to better correlation structures and larger gains.

Formal guarantees. We prove five properties of the CorrCTS family in Appendix D. The most important are: (i) when $\Sigma = I_d$, CorrCTS-Full reduces to standard CTS (Proposition D.1); (ii) Beta posteriors are updated from real rewards only, so the LLM cannot corrupt learning (Proposition D.2); and (iii) in the worst case, RobustCorrCTS incurs at most $O(T/\Delta_{\text{check}})$ excess regret over CTS-Gaussian (Proposition D.3). These guarantees ensure that correlated sampling is a safe augmentation: the algorithm can only help, and degrades gracefully when the LLM is uninformative.

Limitations. All experiments use synthetic Bernoulli rewards, following the standard evaluation protocol for combinatorial semi-bandits (Combes et al., 2015; Wang & Chen, 2018; Kveton et al., 2015). Bernoulli rewards are the canonical test case because they yield Beta posteriors with closed-form updates; validating on a real-world recommendation dataset remains open. We have no formal regret bound for correlated Thompson sampling. A regret bound that captures the dimension reduction from informative covariance structure—analogous to the information gain γ_T in GP-TS (Srinivas et al., 2010)—is an interesting open problem. Only OpenAI models (GPT-4.1-mini, GPT-5-mini, GPT-5.4) were tested as oracles; results with open-source LLMs may differ. The hyperparameters $\rho_{\max} = 0.7$, $\ell = 0.5$, $T_w = 30$ were chosen by preliminary experiments on a separate validation set; a sensitivity analysis is in Appendix A.

7. Conclusion

We showed that using an LLM to derive correlation structure for Thompson sampling is more effective than injecting LLM beliefs into the posterior. Our best algorithm, CorrCTS-Full, reduces cumulative regret by 18.9% relative to standard CTS on combinatorial semi-bandits, with statistical significance confirmed via Holm-Bonferroni correction at $p < 0.0001$ across 320 paired trials. A random-cluster ablation attributes 12.7 percentage points of this gain to the LLM's structural information. All experiments are on synthetic Bernoulli instances with OpenAI models; real-world validation and open-source LLM testing remain open. A concrete theoretical question is whether there exists a regret bound for correlated Thompson sampling that captures the dimension reduction from LLM-derived cluster structure, analogous to the kernel information gain in Gaussian process bandits.

Broader impact. Our work proposes a method for using LLM advice in sequential decision-making. The same technique could improve recommendation or advertising systems; the ethical considerations are those inherent to bandit-driven personalization (filter bubbles, fairness across user groups) rather than anything specific to our method. We do not foresee negative impacts beyond those already

Table 3. Mean cumulative regret by (d, m) setting. Each cell is averaged over 4 gap configurations $\times$ 20 seeds = 80 trials. Best per column in **bold**.

	$d = 25$		$d = 50$			
Algorithm	m=3	m=5	m=3	m=5	vs CTS (avg)	vs Rand (avg)
CorrCTS-Full	**192.0**	**195.1**	**312.5**	**373.3**	+18.9%	+12.7%
RobustCorrCTS	191.3	198.6	351.0	395.8	+14.1%	+7.5%
TS-LLM	214.1	219.8	393.3	543.8	+15.5%	+9.0%
RandomCorr (ablation)	210.9	216.7	386.8	414.4	+7.1%	—
CTS	224.5	223.8	415.7	458.8	—	−7.6%
CUCB	546.4	560.6	933.2	1065.2	−134.8%	—

present in the deployment of combinatorial bandit algorithms.

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A. Full Experimental Details

A.1. Problem Configurations

We use 16 problem configurations, listed in Table 4. For each $(d, m) \in \{25, 50\} \times \{3, 5\}$, we generate 4 instances:

- **Uniform-large**: the top m arms have mean 0.7, the remaining arms have mean $0.7 - \delta$ with $\delta = 0.15$.
- **Uniform-small**: same structure with $\delta = 0.08$.
- **Hard-gap**: the top m arms have mean 0.7; the next m arms have mean $0.7 - 0.05$ (close competitors); remaining arms have mean $0.7 - 0.2$.
- **Staggered**: arms have linearly spaced means from 0.7 down to 0.3, so the gap between consecutive arms is small.

Table 4. Summary of the 16 experimental configurations.

Config	d	m	$\Delta_{\min}$	Type
1–4	25	3	0.15, 0.08, 0.05, varied	Unif-L, Unif-S, Hard, Stag
5–8	25	5	0.15, 0.08, 0.05, varied	Unif-L, Unif-S, Hard, Stag
9–12	50	3	0.15, 0.08, 0.05, varied	Unif-L, Unif-S, Hard, Stag
13–16	50	5	0.15, 0.08, 0.05, varied	Unif-L, Unif-S, Hard, Stag

A.2. Hyperparameters

All algorithms share the following settings:

- Warmup: $T_w = 30$ rounds of standard CTS before the LLM query.
- Horizon: $T = 2500$ rounds.
- LLM model: GPT-5.4 via OpenAI API.
- LLM temperature: 0.3 (for deterministic-ish outputs; model uses reasoning).
- Number of clusters: $K = 8$ for all correlated algorithms.

Algorithm-specific hyperparameters:

- CORRCTS-FULL: $\rho_{\max} = 0.7$, kernel lengthscale $\ell = 0.5$.
- ROBUSTCORRCTS: same kernel parameters, credibility check interval $\Delta_{\text{check}} = 100$, smoothing coefficients $0.7/0.3$.
- RandomCorr: within-cluster correlation $\rho = 0.6$, $K = 8$ (random assignment).
- TS-LLM: decay parameter $\tau = 100$, re-query interval 100 rounds.

A.3. Computational Resources

Experiments were run on a single machine with 16 CPU cores. Each trial (one algorithm, one config, one seed, $T = 2500$) takes approximately 3 seconds for CTS-based algorithms and 5 seconds for TS-LLM (due to more LLM calls). Total wall-clock time for the full experiment: approximately 4 hours. LLM API calls are cached in a local SQLite database; repeated calls with identical prompts are served from cache.

B. Information-Theoretic Analysis of LLM Informativeness

We estimate the mutual information $I(\text{LLM output}; S^\star)$ from cached oracle calls. For each trial, we know the true optimal set $S^\star$ and the LLM's cluster assignment or top-m recommendation. We measure informativeness via the overlap between the LLM's top-m suggestion and $S^\star$.

Let p denote the mean per-arm accuracy (fraction of the LLM's top-m that are truly optimal). The maximum entropy of $S^\star$ for random guessing is $\log_2 \binom{d}{m}$. For $d = 30, m = 5$, this is $\log_2 \binom{30}{5} = 17.12$ bits.

We approximate the conditional entropy as $H(S^\star|\text{LLM}) \approx m \cdot H_{\text{bin}}(p)$ where H_{bin} is the binary entropy function. The estimated mutual information is then $I \approx m \cdot (\log_2(d/m) - H_{\text{bin}}(p))$.

Results across three models:

- **GPT-4.1-mini**: mean overlap 0.78 ($3.9/5$ correct), $I \approx 9.13$ bits (53% of max).
- **GPT-5-mini**: mean overlap 0.80 ($4.0/5$ correct), $I \approx 9.32$ bits (54% of max).
- **GPT-5.4**: mean overlap 0.86 ($4.3/5$ correct), $I \approx 10.36$ bits (60% of max).

The gap between 53% and 60% information efficiency is modest, but the regret difference between GPT-4.1-mini and GPT-5.4 is amplified by the kernel construction: a slightly more accurate cluster assignment can shift inter-cluster correlations substantially.

C. Additional Results

C.1. Holm-Bonferroni Significance Table

Table 5 provides the full Holm-Bonferroni analysis for all algorithms versus CTS.

Table 5. Paired sign test with Holm-Bonferroni correction ($K = 5$, $n = 320$).

Algorithm	W/L	$\bar{\Delta}$	p	Holm α	Sig?
CorrCTS-Full	262/58	62.5	<0.0001	0.0100	Yes
RobustCorrCTS	237/83	46.5	<0.0001	0.0125	Yes
RandomCorr	227/93	23.5	<0.0001	0.0167	Yes
TS-LLM	143/97	10.4	0.0036	0.0250	Yes
CUCB	0/320	−445.6	<0.0001	0.0100	No†

†CUCB is significantly *worse* than CTS.

C.2. CorrCTS-Full vs RobustCorrCTS

CORRCTS-FULL outperforms ROBUSTCORRCTS on average (268.2 vs. 284.2 mean regret), though ROBUSTCORRCTS is competitive at $d = 25$. The credibility-gating interpolation (Equation (3)) introduces additional variance from blending two independent samples, which slightly hurts when the LLM's clusters are already accurate. In practice, CORRCTS-FULL is the simpler and more consistent choice; ROBUSTCORRCTS is preferable in adversarial settings where the LLM oracle is untrusted, since it degrades gracefully to CTS (Proposition D.3).

D. Formal Properties of Correlated Thompson Sampling

We establish five formal properties of the CORRCTS family. Throughout, we write $n_i = \alpha_i + \beta_i$ for the total count of arm i, $\bar{\mu}_i = \alpha_i/n_i$ for the Beta posterior mean, and $\sigma_i^2 = \alpha_i\beta_i/(n_i^2(n_i+1))$ for the Beta posterior variance. We let L denote the lower-triangular Cholesky factor of Σ, so that $LL^\top = \Sigma$.

Proposition D.1 (CTS Recovery)**.** *When $\Sigma = I_d$, Algorithm 1 reduces to standard CTS with Gaussian-approximated posteriors. That is, the sampled parameter vector satisfies $\theta_i \sim \mathcal{N}(\bar{\mu}_i, \sigma_i^2)$ independently across arms $i \in [d]$.*

Proof. When $\Sigma = I_d$, the Cholesky decomposition yields $L = I_d$, since $I_d = I_d \cdot I_d^\top$. At each round $t > T_w$, the algorithm draws $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, I_d)$ and computes $\mathbf{z} = L\boldsymbol{\epsilon} = I_d\boldsymbol{\epsilon} = \boldsymbol{\epsilon}$. Therefore $z_i = \epsilon_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$ for each $i \in [d]$.

Line 14 of Algorithm 1 sets $\theta_i = \bar{\mu}_i + \sigma_i \cdot z_i = \bar{\mu}_i + \sigma_i \cdot \epsilon_i$. Since $\epsilon_i \sim \mathcal{N}(0,1)$ independently, we have $\theta_i \sim \mathcal{N}(\bar{\mu}_i, \sigma_i^2)$ independently for all i. This is exactly the Gaussian moment-matched approximation to $\text{Beta}(\alpha_i, \beta_i)$, with each arm sampled independently—i.e., standard CTS with Gaussian sampling. □

Proposition D.2 (Posterior Integrity)**.** *The Beta posteriors $(\alpha_i, \beta_i)_{i\in[d]}$ in Algorithm 1 are updated identically to standard CTS. The LLM-derived covariance matrix Σ affects only the sampling distribution, not the posterior update rule.*

Proof. We exhibit the separation between the two operations in Algorithm 1.

Posterior update (lines 16–17). After playing S_t and observing $(X_{i,t})_{i\in S_t}$, the update rule is:

$$\alpha_i \leftarrow \alpha_i + X_{i,t}, \quad \beta_i \leftarrow \beta_i + (1 - X_{i,t}), \quad \forall\, i \in S_t.$$

This is the standard Beta-Bernoulli conjugate update. Crucially, the right-hand side depends only on the current (α_i, β_i) and the observed reward $X_{i,t} \in \{0,1\}$. Neither Σ, L, $\boldsymbol{\epsilon}$, nor any LLM output enters this computation.

Sampling step (lines 11–14). The correlated sample $\theta_i = \bar{\mu}_i + \sigma_i \cdot (L\boldsymbol{\epsilon})_i$ reads (α_i, β_i) to compute $(\bar{\mu}_i, \sigma_i)$ and uses Σ through its Cholesky factor L. This step produces the decision variable $\boldsymbol{\theta}$ but does not write to (α_i, β_i).

Since the posterior update at time t depends only on $(\alpha_i^{(t)}, \beta_i^{(t)}, X_{i,t})$ and the sampling step depends only on $(\alpha_i^{(t)}, \beta_i^{(t)}, \Sigma)$, with neither modifying the other's inputs, the two operations are functionally independent. By induction on t, the sequence $\{(\alpha_i^{(t)}, \beta_i^{(t)})\}_{t\geq 1}$ is identical to that of standard CTS run on the same reward sequence. □

Proposition D.3 (Worst-Case Guarantee for ROBUSTCORRCTS)**.** *For any LLM oracle and any problem instance, the expected regret of* ROBUSTCORRCTS *(Algorithm 2) satisfies*

$$\mathbb{E}[R_T(\textsc{RobustCorrCTS})] \;\leq\; \mathbb{E}[R_T(\textit{CTS-Gaussian})] \;+\; O\Big(\frac{T}{\Delta_{\text{check}}}\Big),$$

where CTS-Gaussian denotes CTS with independent Gaussian-approximated posteriors, and the additive term accounts for the credibility check overhead.

Proof. Consider the worst case for the LLM oracle: the clusters returned are adversarial, so the overlap between the LLM's top cluster and the empirically best arms is consistently zero. We analyze the credibility weight and the sampling distribution.

Step 1: Credibility decay. The credibility weight evolves as $w_{t+\Delta_{\text{check}}} = 0.7\, w_t + 0.3\,\text{overlap}_t$. When $\text{overlap}_t = 0$ at every check, we have $w_t = 0.7^{\lfloor (t-T_w)/\Delta_{\text{check}} \rfloor}$. After k checks, $w_t = 0.7^k$. For $k \geq k_0 := \lceil \log(1/\delta)/\log(10/7) \rceil$, we have $w_t \leq \delta$ for any desired $\delta > 0$.

Step 2: Sampling distribution. At round t, Algorithm 2 produces $\boldsymbol{\theta}_t = w_t \boldsymbol{\theta}^{\text{corr}} + (1 - w_t)\boldsymbol{\theta}^{\text{CTS}}$, where $\boldsymbol{\theta}_i^{\text{CTS}} = \bar{\mu}_i + \sigma_i \epsilon_{2,i}$ with $\epsilon_{2,i} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$. As $w_t \to 0$, we have $\boldsymbol{\theta}_t \to \boldsymbol{\theta}^{\text{CTS}}$, which is exactly CTS-Gaussian sampling. More precisely, $\theta_{t,i} = \bar{\mu}_i + \sigma_i[w_t(L\boldsymbol{\epsilon}_1)_i + (1 - w_t)\epsilon_{2,i}]$. Since $\boldsymbol{\epsilon}_1, \boldsymbol{\epsilon}_2$ are independent, $\theta_{t,i}$ is Gaussian with mean $\bar{\mu}_i$ and variance $\sigma_i^2[w_t^2(LL^\top)_{ii} + (1 - w_t)^2 + 2w_t(1 - w_t) \cdot 0] = \sigma_i^2[w_t^2 \Sigma_{ii} + (1 - w_t)^2]$. Since $\Sigma_{ii} = 1$, this variance equals $\sigma_i^2[w_t^2 + (1 - w_t)^2] \leq \sigma_i^2$ for all $w_t \in [0,1]$, with equality at $w_t \in \{0,1\}$.

Step 3: Regret decomposition. Partition the horizon into three phases: (i) warmup rounds $t \leq T_w$: standard CTS, contributing regret $\leq \mathbb{E}[R_{T_w}(\text{CTS})]$; (ii) credibility check rounds: at most $\lfloor T/\Delta_{\text{check}} \rfloor$ checks, each incurring at most $\Delta_{\max} = \max_i \mu_i - \min_i \mu_i \leq 1$ instantaneous regret for the comparison computation, contributing $O(T/\Delta_{\text{check}})$; (iii) all other rounds $t > T_w$: once $w_t \leq \delta$, the marginal distribution of each $\theta_{t,i}$ is within total variation distance $O(\delta)$ of $\mathcal{N}(\bar{\mu}_i, \sigma_i^2)$. By a coupling argument, the actions chosen by ROBUSTCORRCTS and CTS-Gaussian differ with probability at most $O(d\delta)$ per round. Choosing $\delta = 1/T$ and noting that the convergence takes $O(\log T)$ checks, the excess regret from the transient phase is $O(\Delta_{\text{check}} \log T)$. Combining: $\mathbb{E}[R_T(\textsc{RobustCorrCTS})] \leq \mathbb{E}[R_T(\text{CTS-Gaussian})] + O(T/\Delta_{\text{check}})$. □

Proposition D.4 (Gaussian Approximation Validity)**.** *Let $X \sim \text{Beta}(\alpha, \beta)$ with $n = \alpha + \beta \geq 2$, $\mu = \alpha/n$, and $\sigma^2 = \alpha\beta/(n^2(n+1))$. Let $Y \sim \mathcal{N}(\mu, \sigma^2)$. Then:*

(a) The first two moments match exactly: $\mathbb{E}[X] = \mathbb{E}[Y] = \mu$ and $\text{Var}(X) = \text{Var}(Y) = \sigma^2$.

(b) The total variation distance satisfies $d_{\text{TV}}(\text{Beta}(\alpha, \beta), \mathcal{N}(\mu, \sigma^2)) \leq C/\sqrt{n}$ for a universal constant $C > 0$.

Proof. Part (a). By definition, $\mathbb{E}[X] = \alpha/(\alpha+\beta) = \mu$ and $\mathrm{Var}(X) = \alpha\beta/[(\alpha+\beta)^2(\alpha+\beta+1)] = \sigma^2$. The Gaussian $Y \sim \mathcal{N}(\mu, \sigma^2)$ has $\mathbb{E}[Y] = \mu$ and $\mathrm{Var}(Y) = \sigma^2$ by construction. Hence the first two moments match exactly.

Part (b). We use the Pólya urn representation. If $U_1, \ldots, U_n$ are the order statistics of $n-1$ i.i.d. $\mathrm{Uniform}(0,1)$ random variables, then $U_{(\alpha)} \sim \mathrm{Beta}(\alpha, \beta)$ where $\alpha + \beta = n$. Equivalently, $X \overset{d}{=} \sum_{j=1}^{\alpha} Z_j/n$ where $(Z_1, \ldots, Z_n)$ are negatively associated indicator random variables with $\sum Z_j = \alpha$ marginally. By the representation $X = \bar{Z}$ (a normalized sum), we apply the Berry–Esseen theorem for the standardized variable $(X-\mu)/\sigma$. The Berry–Esseen bound for sums of bounded random variables gives

$$\sup_{x\in\mathbb{R}} \big|\mathbb{P}\big((X-\mu)/\sigma \le x\big) - \Phi(x)\big| \;\le\; \frac{C_0}{\sqrt{n}},$$

where C_0 is a universal constant and Φ is the standard normal CDF. This is a bound on the Kolmogorov distance d_K. Since for absolutely continuous distributions the total variation distance satisfies $d_{\mathrm{TV}} \le 2\sqrt{d_K}$ (by the Devroye–Györfi inequality between variational and Kolmogorov distances for unimodal densities), we obtain $d_{\mathrm{TV}}(\mathrm{Beta}(\alpha,\beta), \mathcal{N}(\mu,\sigma^2)) \le C/\sqrt{n}$ for $C = 2\sqrt{C_0}$. Alternatively, direct Fourier analysis of the Beta characteristic function (which is a ratio of gamma functions) yields the sharper bound $d_{\mathrm{TV}} \le C'/n$ for balanced α, β; the $O(1/\sqrt{n})$ bound holds uniformly over all $\alpha, \beta \ge 1$ with $\alpha + \beta = n$. □

Proposition D.5 (Positive Definiteness of Σ)**.** *Let $f_1, \ldots, f_d \in \mathbb{R}$ be the cluster-rank features, $\rho_{\max} \in (0,1)$, and $\ell > 0$. Define the $d \times d$ matrix Σ by $\Sigma_{ij} = \rho_{\max} \cdot \exp\big(-(f_i - f_j)^2/(2\ell^2)\big)$ for $i \ne j$ and $\Sigma_{ii} = 1$. Then Σ is positive definite.*

Proof. Define the RBF (squared exponential) kernel $k(x,y) = \exp\big(-\|x-y\|^2/(2\ell^2)\big)$ for $x, y \in \mathbb{R}$. Let $K \in \mathbb{R}^{d\times d}$ be the Gram matrix with $K_{ij} = k(f_i, f_j)$. Since k is a Mercer kernel—it is the reproducing kernel of the RKHS of smooth functions on $\mathbb{R}$ (Srinivas et al., 2010)—the matrix K is positive semi-definite. Moreover, $K_{ii} = 1$ for all i. If the features $f_1, \ldots, f_d$ are not all identical, K is strictly positive definite (the RBF kernel is *universal*, i.e., its Gram matrix on distinct points is positive definite).

Now decompose Σ:

$$\Sigma \;=\; \rho_{\max} \cdot K \;+\; (1-\rho_{\max}) \cdot I_d.$$

To verify: for $i \ne j$, $[\rho_{\max} K + (1-\rho_{\max}) I]_{ij} = \rho_{\max} K_{ij} + 0 = \rho_{\max} \exp(-(f_i - f_j)^2/(2\ell^2)) = \Sigma_{ij}$. For $i = j$, $\rho_{\max} \cdot 1 + (1-\rho_{\max}) \cdot 1 = 1 = \Sigma_{ii}$.

Since $K \succeq 0$ and $I_d \succ 0$, and $\rho_{\max} \in (0,1)$, we have

$$\Sigma = \rho_{\max} K + (1-\rho_{\max}) I_d \;\succeq\; (1-\rho_{\max}) I_d \;\succ\; 0.$$

Hence Σ is positive definite, with minimum eigenvalue at least $1 - \rho_{\max} > 0$. In particular, the Cholesky factorization $\Sigma = LL^\top$ exists and is unique. □